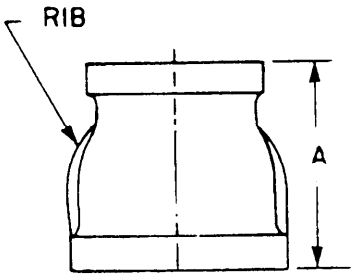
MILITARY SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; REDUCER, PIPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

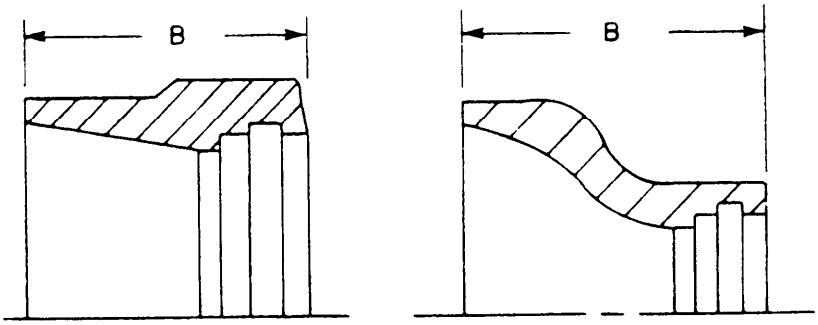
The requirements for acquiring the reducers described herein shall consist of this specification and the latest issue of MIL-F-1183.



(Commercial name - reducing coupling)

SH 12174

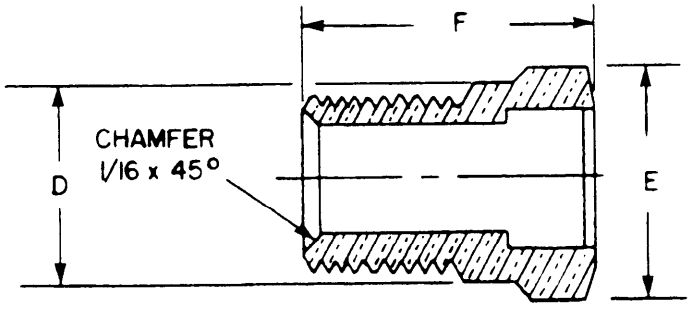
FIGURE 1. Reducer (socket to socket).



(Commercial name - bushing)

SH 12175

FIGURE 2. Reducer (male pipe to socket).



(Commercial name - adapter)

SH 12176

FIGURE 3. Reducer (male pipe thread to socket).

AMSC N/A FSC 4730

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TABLE I. Dimensions.

Nominal pipe size of larger socket (Figure 1)	Nominal pipe size of male pipe end (Figure 2)	Nominal pipe size male pipe thread end (Figure 3)	Nominal pipe size reduced socket end (Fig. 1, 2 or 3)	Female reducer length (Inches) A	Male to female reducer length (Inches) B	Male pipe thread to socket reducer		
						Minimum diameter (Inches) D	Minimum band dia. (Inches) E	Length (Inches) F
3/8	3/8	3/8	1/4	1.01	0.94	0.75	0.81	1.06
1/2	1/2		1/4	1.13	1.06			
1/2	1/2	1/2	3/8	1.17	1.06	.91	1.00	1.31
3/4	3/4		3/8	1.24	1.16			
3/4	3/4	3/4	1/2	1.36	1.16	1.09	1.19	1.38
1	1		1/2	1.49	1.22			
1	1	1	3/4	1.56	1.22	1.31	1.44	1.63
1-1/4	1-1/4		3/4	1.65	1.44			
1-1/4	1-1/4	1-1/4	1	1.77	1.44	1.63	1.75	1.72
1-1/2	1-1/2		1	1.80	1.63			
1-1/2	1-1/2	1-1/2	1-1/4	1.89	1.63	2.00	2.11	1.88
2	2		1-1/4	2.03	1.75			
2	2	2	1-1/2	2.06	1.75	2.25	2.50	1.97
2-1/2	2-1/2		1-1/2	3.25	1.94			
2-1/2	2-1/2	2-1/2	2	3.25	1.94	2.81	3.03	2.63
3	3		2	3.69	2.13			
3	3		2-1/2	3.69	2.13			
3-1/2	3-1/2		2-1/2	4.00	2.19			
3-1/2	3-1/2		3	4.00	2.19			
4	4		3	4.38	2.25			
4	4		3-1/2	4.38	2.25			
5	5		3-1/2	5.00	2.75			
5	5		4	5.00	2.75			
6			4	5.75				
6			5	5.75				

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REQUIREMENTS:

1. Socket-to-socket reducer type. Socket-to-socket reducers (see figure 1) except the 6 by 5-inch reducer shall be type A (streamlined interior). The 6 by 5-inch reducer is a type B (interior not streamlined).
2. Male pipe-to-socket reducer type. Male pipe-to-socket reducers (see figure 2) shall be type A (streamlined interior) which is achieved by tapering the interior to provide a gradual transition from the large diameter to the small diameter. Either method of transition specified on figure 2 is acceptable provided the body wall thickness is not less than that specified in table V of MIL-F-1183 for the required end sizes.
3. Male pipe thread-to-socket reducer type. Male pipe thread-to-socket reducers (see figure 3) shall be type B (interior not streamlined).
4. Dimensions. Dimensions for reducers (see figures 1, 2 and 3) shall be as specified in table I. The tolerance for the end-to-end (length) dimensions shall be twice the tolerance specified in table IV of MIL-F-1183 for the nominal pipe size of the reduced socket end.
5. Ribs. Reducers may be furnished with or without ribs.
6. Specification sheet part number. Reducers covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-1183).

M1183/8 - XX X

Specification sheet code number
Reducer style and pipe size code number (see requirement 7 herein)
Brazing ring groove and brazing ring code letter (see requirement 8 herein)

7. Reducer style and pipe size. The reducer style (socket-to-socket, male pipe-to-socket and male pipe thread-to-socket) and nominal pipe sizes are identified by two digits (see table II). NOTE: Socket-to-socket reducer has an industry nomenclature of reducing coupling, male pipe-to-socket reducer has an industry nomenclature of bushing and male pipe thread-to-socket reducer has an industry nomenclature of adapter.

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TABLE II. Code number to style and size.

Reducer style			Nominal pipe size (Inches)
1/ Socket-to-socket (figure 1)	2/ Male pipe-to-socket (figure 2)	3/ Male pipe thread-to-socket (figure 3)	
01	41	81	3/8 x 1/4
02	42		1/2 x 1/4
03	43	82	1/2 x 3/8
04	44		3/4 x 3/8
05	45	83	3/4 x 1/2
06	46		1 x 1/2
07	47	84	1 x 3/4
08	48		1-1/4 x 3/4
09	49	85	1-1/4 x 1
10	50		1-1/2 x 1
11	51	86	1-1/2 x 1-1/4
12	52		2 x 1-1/4
13	53	87	2 x 1-1/2
14	54		2-1/2 x 1-1/2
15	55	88	2-1/2 x 2
16	56		3 x 2
17	57		3 x 2-1/2
18	58		3-1/2 x 2-1/2
19	59		3-1/2 x 3
20	60		4 x 3
21	61		4 x 3-1/2
22	62		5 x 3-1/2
23	63		5 x 4
24			6 x 4
25			6 x 5

1/ Type A except 6 by 5-inch size (see requirement 1).
2/ Type A (see requirement 2).
3/ Type B (see requirement 3).

8. Brazing ring groove and brazing ring. Face fed (no groove), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the reducer or grooved with brazing rings separately packaged and shipped with but not attached to the reducer socket ends are identified by a single letter (see table III).

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TABLE III. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/	Grooved with brazing ring separately packaged and shipped with

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for elbow with the code letter of G, R or S.

2/ When the socket condition code letter is A, the brazing rings shall not be inserted in the grooves but shall be packaged separately and either included within each individual reducer package or physically attached to the reducer.

9. Specification part number definition. The specification sheet part number is a definitive part number which is formed by combining the specification sheet code number (M1183/8 for MIL-F-1183/8) with a dash after it, the reducer style and pipe size code number and the brazing ring groove and brazing ring code letter as specified in requirement 6.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Navy - SH

Agent:
DLA - CS

(Project 4730-0757)

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